

# One specialized MCP server. Many AI clients. One SAP context.

VERTEX plugins bring SAP development into the tools developers already use — with a secure, action-ready workflow.

VERTEX plugins

Multi-AI enabled

Specialized MCP server

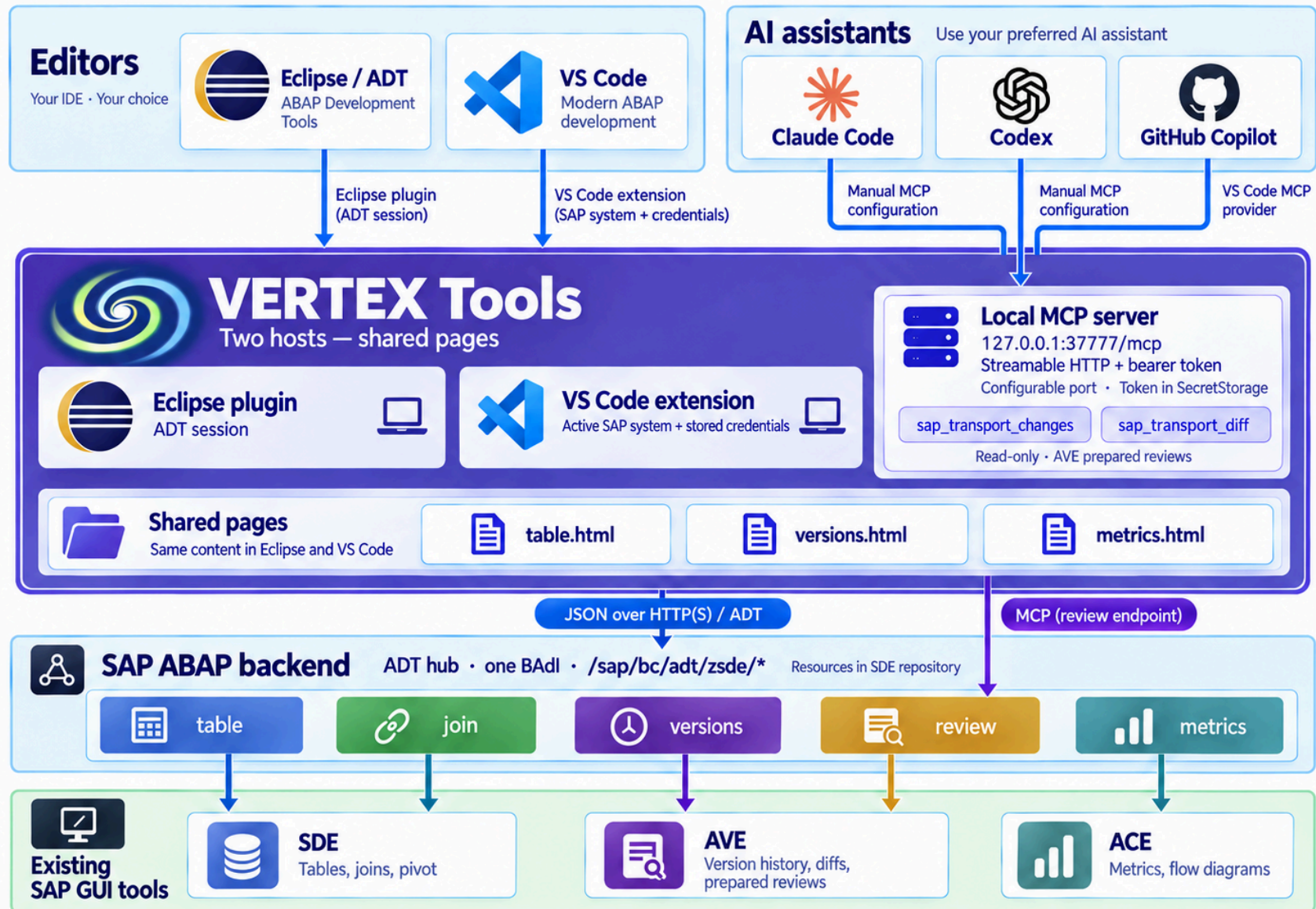
SAP development context

Marketplace: Visual Studio Marketplace

GitHub: [github.com/ysichov/VERTEX](https://github.com/ysichov/VERTEX)

# VERTEX Architecture

ABAP tools for Eclipse, VS Code and AI assistants



AI assistants use their own models and subscriptions. VERTEX provides SAP context.

Versions

Review

← Review EXECUTE\_SQL 00001 → active +20 -7 Full

50 IF i\_quiet = abap\_false.

#2 added, 2 lines · line 51 · YSYCHOV · open 

Approve Decline Comment

- MESSAGE 'Statement must start with SELECT' TYPE 'S' DISPLAY LIKE 'E'.

51 + MESSAGE ev\_error TYPE 'S' DISPLAY LIKE 'E'.

52 ENDIF.

53 RETURN.

54 ENDIF.

55 FIND REGEX '\sFROM\s' IN l\_upper MATCH OFFSET DATA(1\_from\_off) MATCH LENGTH DATA(1\_from\_len).

56 IF sy-subrc NE 0.

#3 added, 1 line · line 57 · YSYCHOV · open 

Approve Decline Comment

57 + ev\_error = 'FROM clause not found'.

58 IF i\_quiet = abap\_false.

#4 added, 2 lines · line 59 · YSYCHOV · open 

Approve Decline Comment

- MESSAGE 'FROM clause not found' TYPE 'S' DISPLAY LIKE 'E'.

59 + MESSAGE ev\_error TYPE 'S' DISPLAY LIKE 'E'.

60 ENDIF.

61 RETURN.

62 ENDIF.

:

108 FIND REGEX '(\w+)~(\w+)' IN l\_fld\_up SUBMATCHES l\_alias2 l\_field.

109 ENDIF.

110 IF l\_agg IS INITIAL OR l\_name IS INITIAL OR l\_field IS INITIAL.

#5 added, 1 line · line 111 · YSYCHOV · open 

Approve Decline Comment

111 + ev\_error = |Cannot parse field: { l\_fld\_str } (expressions need AS name)|.

112 IF i\_quiet = abap\_false.

#6 added, 2 lines · line 113 · YSYCHOV · open 

Approve Decline Comment

- MESSAGE |Cannot parse field: { l\_fld\_str } (expressions need AS name)| TYPE 'S' DISPLAY LIKE 'E'.

113 + MESSAGE ev\_error TYPE 'S' DISPLAY LIKE 'E'.

114 ENDIF.

115 RETURN.

116 ENDIF.

:

161 RECEIVING p\_descr\_ref = lo\_td

162 EXCEPTIONS type\_not\_found = 1 OTHERS = 2 ).

163 IF sy-subrc NE 0 OR lo\_td IS NOT BOUND.

Assistant

Codex gpt-5.6-luna (config.toml)

General review

General review

SQL and performance

Security

Tests

New conversation

or handling  
ort concrete  
tions, impact  
and suggested fixes. Answer in the  
user's language.

Save profile

Delete profile

You

Please review E19K907006

Codex · gpt-5.6-luna · E19

Review of transport E19K907006 found these issues:

1. High — raw SQL injection through join conditions. 'GET' accepts 'onTn' directly (ZCL\_SDE\_ADT\_RES\_JOIN, GET lines 83–97), and 'SET\_JOIN' stores it after only whitespace normalization (ZCL\_SDE\_TOOLS, SET\_JOIN lines 7–12). A caller can inject arbitrary SQL predicates or syntax into the generated JOIN. Parse and whitelist aliases, fields, operators, and literals; preferably construct the condition from validated field selections rather than accepting SQL text

For example: the last change of  
ZCL\_AVE\_POPUP=>BUILD\_LAYOUT



> Table SFLIGHT Load Join Open in new window Assistant

94 rows

Assistant

Codex gpt-5.6-luna (config.toml)

You

Please create a join of SFLIGHT with SCARR and spfli . Only key fields + company name

Codex · gpt-5.6-luna · E19

Set up the SFLIGHT join with SPFLI and SCARR, selecting only key fields and the airline company name.

SFLIGHT · join SPFLI, SCARR · 7 fields

| t0_carrid | t0_connid | t0_fldate  | t1_carrid | t1_connid | t2_carrid | t2_carrname       |
|-----------|-----------|------------|-----------|-----------|-----------|-------------------|
| AA        | 17        | 2026-01-13 | AA        | 17        | AA        | American Airlines |
| AA        | 17        | 2026-04-03 | AA        | 17        | AA        | American Airlines |
| AA        | 17        | 2026-06-22 | AA        | 17        | AA        | American Airlines |
| AA        | 17        | 2026-09-10 | AA        | 17        | AA        | American Airlines |
| AA        | 17        | 2026-11-29 | AA        | 17        | AA        | American Airlines |
| AA        | 17        | 2027-02-17 | AA        | 17        | AA        | American Airlines |
| AZ        | 555       | 2026-01-13 | AZ        | 555       | AZ        | Alitalia          |
| Δ7        | 555       | 2026-04-03 | Δ7        | 555       | Δ7        | Alitalia          |

## Tables around sflight

→ T000 → SPFLI → SAPLANE → SCURX → SCARR ← SBOOK ← SNVOICE ← STICKET → SAIRPORT → SGEOCITY  
← DAAG\_SBOOK ← DAAG\_SPFLI ← SCARPLAN ← SCOUNTER ← SDESSERT ← SMACOURSE ← SMEAL ← SMEALT  
← SSTARTER

## The join

| Alias | Table                 | Join       | On                                              |
|-------|-----------------------|------------|-------------------------------------------------|
| T0    | SFLIGHT               |            |                                                 |
| T1    | SPFLI Flight schedule | LEFT OUTER | t1~CARRID = t0~CARRID AND t1~CONNID = t0~CONNID |
| T2    | SCARR Airline         | LEFT OUTER | t2~CARRID = t0~CARRID                           |

## The SELECT list

Everything all none keys

T0 SFLIGHT all none keys

☒ carrid ☒ connid ☒ fldate ☐ price ☐ currency ☐ planetype ☐ seatsmax ☐ seatsocc ☐ paymentsum  
☐ seatsmax\_b ☐ seatsocc\_b ☐ seatsmax\_f ☐ seatsocc\_f

T1 SPFLI all none keys

For example: SFLIGHT for carrier AA, joined with